

# Evaluating the Cost-Effectiveness Of *Chlorella vulgaris* Biodiesel Cultivation Methods

## ABSTRACT

- Global warming, driven by increased greenhouse gas emissions from fossil fuel use, remains a **major scientific and environmental challenge**. Furthermore, **diminishing quantities and price fluctuation** lower the overall **attainability and future sustainability** of fossil fuels. In response, researchers have **explored alternative sustainable energy sources** including **biofuels**, which convert **triglycerides** in organic feedstocks into usable **hydrocarbon fuels** through **transesterification**.

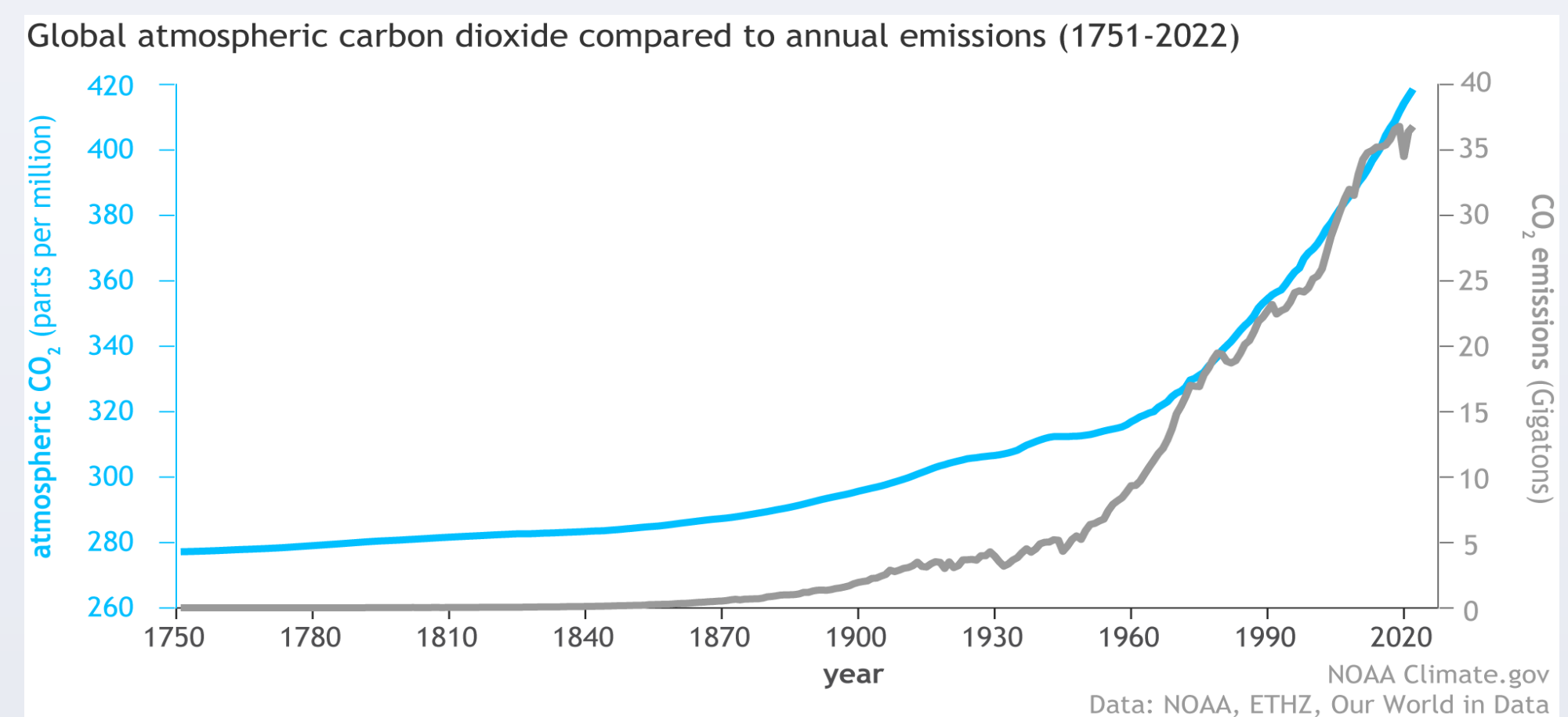


Figure 1. Atmospheric carbon dioxide versus annual emissions (National Oceanic and Atmospheric Administration).

- Algae** has emerged as a **promising feedstock** due to its high biomass yield, high energy content, and ability to grow in a wide range of water conditions. However, **current algae cultivation methods remain costly**, limiting their economic feasibility.
- Photobioreactors (PBR)** Utilize artificial light and controlled conditions to produce high quality fuel. **Open pond reactors (OPR)** rely on natural sunlight and are significantly less expensive but produce lower quality fuel.
- This study evaluates **the cost-effectiveness** of these two methodologies.

## OBJECTIVES

- The objective for this experiment was to determine **which of the two primary algae cultivation methods**, photobioreactors (PBR) or open pond reactors (OPR), is more cost effective for biofuel production. While previous research has compared these two methods, **few studies have evaluated their economic feasibility** based on energy output.
- To evaluate this, **biodiesel** was produced from **two algae colonies** grown under **different conditions**: one using **artificial growth light** (PBR conditions) and the other using **natural light** (OPR conditions). The **calorific value** of each fuel was **then measured** to assess the **energy output** of both fuels.

## HYPOTHESIS

- The **independent variable** was the **cultivation method**, and our **dependent variable** was the **calorific value** of the resulting biodiesel.
- The **null hypothesis** stated that the calorific value of **PBR-derived** biodiesel **would not be at least 30%** greater than that of **OPR-derived** biofuel and therefore would **not justify** the increased cost. The **alternative hypothesis** stated that the calorific value of **PBR-derived** biodiesel **would be at least 30%** greater than that of OPR-derived biofuel, **justifying** the higher cost.
- A **plain cotton ball** was utilized as a control.

## MATERIALS AND METHODS

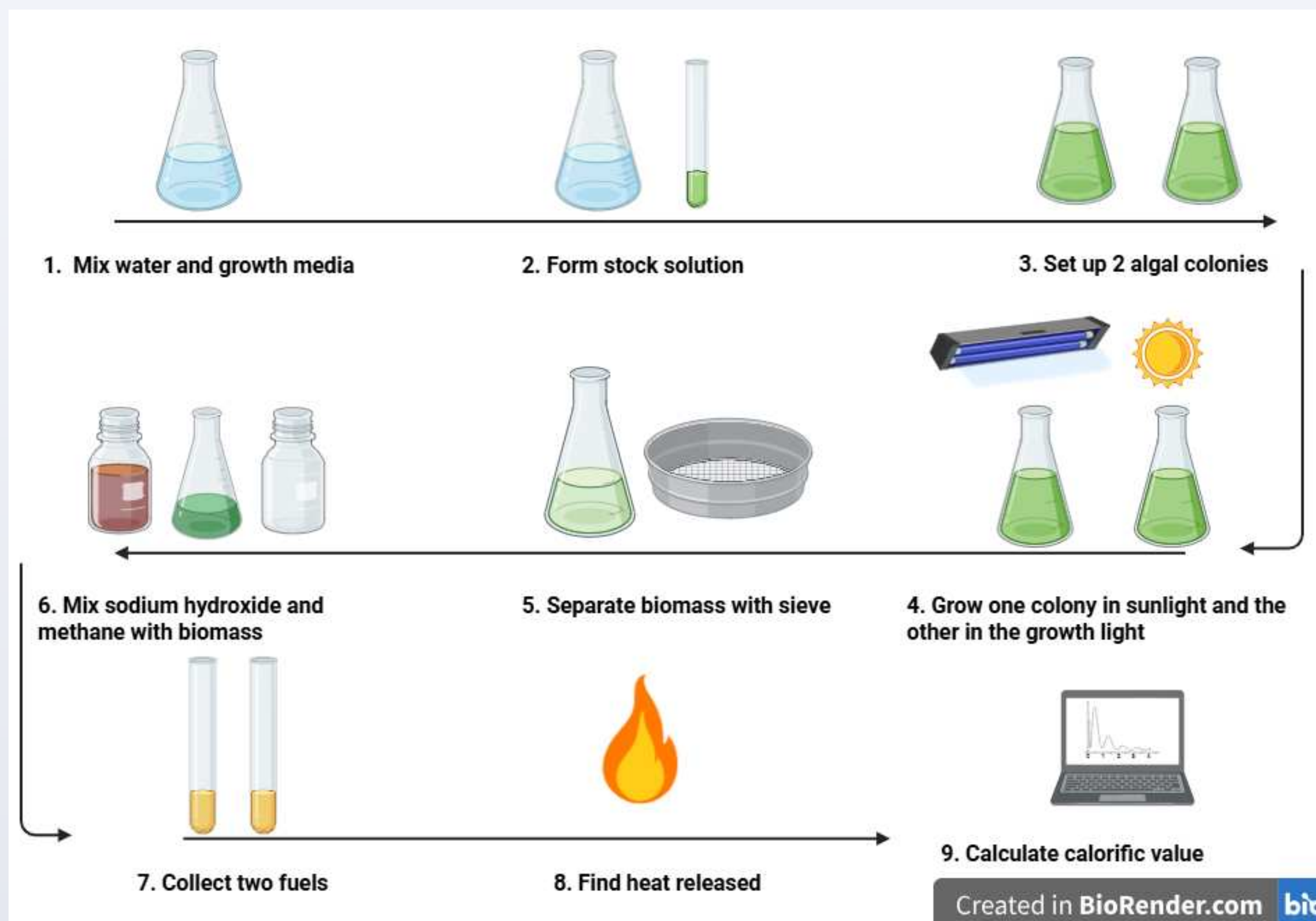


Figure 2. Experimental design (Created in BioRender).



Figure 3. Calorimeter used for data collection.

## RESULTS

| Cultivation method:     | Mass (Grams) | Change In Temperature (Celsius) | Specific Heat Capacity of Water (Constant) (J/g·°C) | Water Mass (g) | Heat Released (J) | Calorific Value (J/G) |
|-------------------------|--------------|---------------------------------|-----------------------------------------------------|----------------|-------------------|-----------------------|
| Photobioreactor         | 1.9473       | 18                              | 4.186                                               | 125            | 9418.5            | 4836.696965           |
| Open Pond reactor       | 0.7347       | 7                               | 4.186                                               | 125            | 3139.5            | 4985.368177           |
| N/A (Plain cotton ball) | 0.639        | 5                               | 4.186                                               | 125            | 2616.25           | 4120.07874            |

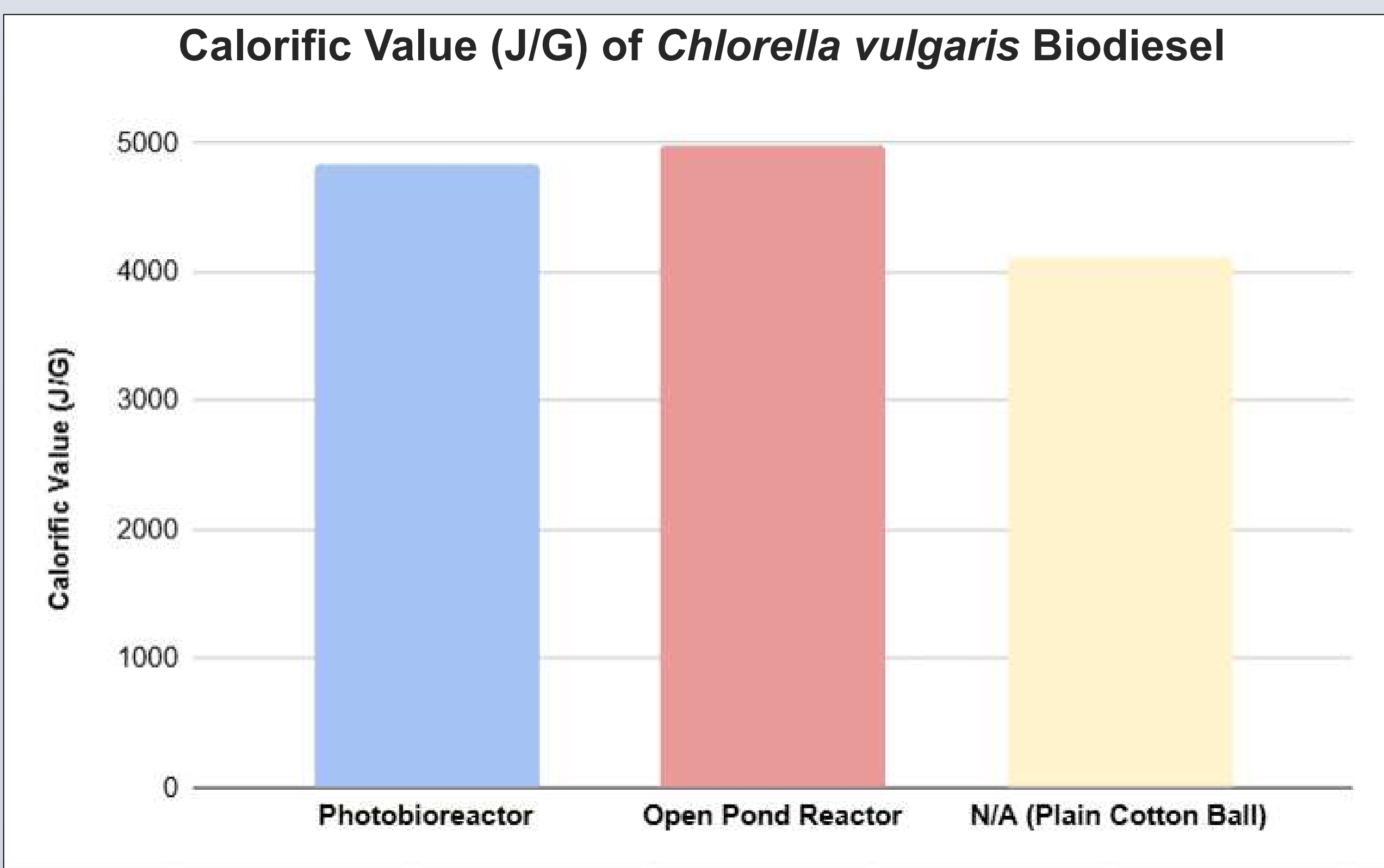


Figure 4. Processed calorific value of *Chlorella vulgaris* biodiesel.

- Raw data** included the **mass of the cotton ball** with fuel, the **initial water temperature** before combustion, and the **final water temperature** after combustion.
- The **change in water temperature** was **calculated** to determine the **heat released** by each fuel sample.
- This **heat released** was then divided by the **mass of the fuel burned** to calculate the **calorific value**, representing the **energy released per gram** of fuel combusted.



Figure 5. PBR colony *Chlorella vulgaris* biodiesel.

## CONCLUSION

- The open pond reactor (OPR) biodiesel **produced a higher calorific value** than the photobioreactor (PBR) fuel, **contradicting** previous research. Given its lower initial cost and comparable energy output, **OPR can be considered the more cost-effective cultivation method** for *Chlorella vulgaris*-based biodiesel, and the **null hypothesis was not rejected**.
- Future improvements to this experiment potentially include **increasing the number of trials** and producing **larger quantities of biodiesel** to improve reliability and confidence in the results.
- Determining the more optimal cultivation method for algae-based biofuels will **encourage future research** to explore ways of **lowering costs** and **increasing feasibility**, bringing us one step closer to a carbon neutral world.
- As greenhouse gas emissions rise and fossil fuel costs increase due to global instability, this **study highlights algae-based biodiesel as a near carbon-neutral and resilient alternative** that can be implemented at scale today.

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